

Table 1 continued

Saturn

$$\alpha_0 = 40.589 - 0.036T$$

$$\delta_0 = 83.537 - 0.004T$$

$$W = 38.90 + 810.7939024d \quad (e)$$

Uranus

$$\alpha_0 = 257.311$$

$$\delta_0 = -15.175$$

$$W = 203.81 - 501.1600928d \quad (e)$$

Neptune

$$\alpha_0 = 299.36 + 0.70 \sin N$$

$$\delta_0 = 43.46 - 0.51 \cos N$$

$$W = 253.18 + 536.3128492d - 0.48 \sin N \quad (e)$$

$$N = 357.85 + 52.316T$$

(a) The equation W for the Sun is now corrected for light travel time and removing the aberration correction. See the Appendix in [Seidelmann et al. \(2007\)](#)

(b) The 20° meridian is defined by the crater Hun Kal

(c) The 0° meridian is defined by the central peak in the crater Ariadne

(d) The 0° meridian is defined by the crater Airy-0

(e) The equations for W for Jupiter, Saturn, Uranus and Neptune refer to the rotation of their magnetic fields (System III). On Jupiter, System I ($W_I = 67^\circ.1 + 877^\circ.900d$) refers to the mean atmospheric equatorial rotation; System II ($W_{II} = 43^\circ.3 + 870^\circ.270d$) refers to the mean atmospheric rotation north of the south component of the north equatorial belt, and south of the north component of the south equatorial belt

Table 2 Recommended values for the direction of the north pole of rotation and the prime meridian of the satellites

α_0 , δ_0 , T , and d have the same meanings as in Table 1 (epoch JD 2451545.0, i.e. 2000 January 1 12 hours TDB)

Earth:

$$\begin{aligned} \text{Moon} \quad \alpha_0 = & 269^\circ.9949 + 0^\circ.0031T - 3^\circ.8787 \sin E1 - 0^\circ.1204 \sin E2 \\ & + 0.0700 \sin E3 - 0.0172 \sin E4 + 0.0072 \sin E6 \\ & - 0.0052 \sin E10 + 0.0043 \sin E13 \end{aligned} \quad (a)$$

$$\begin{aligned} \delta_0 = & 66.5392 + 0.0130T + 1.5419 \cos E1 + 0.0239 \cos E2 \\ & - 0.0278 \cos E3 + 0.0068 \cos E4 - 0.0029 \cos E6 \\ & + 0.0009 \cos E7 + 0.0008 \cos E10 - 0.0009 \cos E13 \end{aligned}$$

$$\begin{aligned} W = & 38.3213 + 13.17635815d - 1.4 \times 10^{-12}d^2 + 3.5610 \sin E1 \\ & + 0.1208 \sin E2 - 0.0642 \sin E3 + 0.0158 \sin E4 \\ & + 0.0252 \sin E5 - 0.0066 \sin E6 - 0.0047 \sin E7 \\ & - 0.0046 \sin E8 + 0.0028 \sin E9 + 0.0052 \sin E10 \\ & + 0.0040 \sin E11 + 0.0019 \sin E12 - 0.0044 \sin E13 \end{aligned}$$

where

$$\begin{aligned} E1 = & 125^\circ.045 - 0^\circ.0529921d, \quad E2 = 250^\circ.089 - 0^\circ.1059842d, \quad E3 = 260^\circ.008 + 13^\circ.0120009d, \\ E4 = & 176.625 + 13.3407154d, \quad E5 = 357.529 + 0.9856003d, \quad E6 = 311.589 + 26.4057084d, \\ E7 = & 134.963 + 13.0649930d, \quad E8 = 276.617 + 0.3287146d, \quad E9 = 34.226 + 1.7484877d, \\ E10 = & 15.134 - 0.1589763d, \quad E11 = 119.743 + 0.0036096d, \quad E12 = 239.961 + 0.1643573d, \\ E13 = & 25.053 + 12.9590088d \end{aligned}$$